Jayanth Dasamantharao

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WORK EXPERIENCE

AI Engineer - Fedway Associates, *EliteUS Software Solutions* — *July 2024 - Present | Edison, NJ*

- Built agentic AI systems using LangChain, AutoGen, and Azure Search for real-time decision support tools.
- Developed production-grade RAG pipelines (OpenAI, LLaMA, Titan) improving contextual search precision.
- Deployed MLOps pipelines using Apache Kafka and AWS, reducing model delivery cycle time by 40%.
- Led integration of AI systems into full-stack applications (React, Flask), enabling scalable user interfaces.
- Presented data products and pipelines to cross-functional teams, improving adoption and feature planning.

Data Science Intern, *Harvest Software Solutions* — *June 2023 - Sept 2023 | Jacksonville, Florida*

- Forecasted ad pricing using XGBoost & LightGBM increasing pricing accuracy and improving ad revenue strategy.
- Built ETL pipelines with SQLAlchemy, Pandas to support pricing analytics and experimentation frameworks.
- Created NLP models with Hugging Face Transformers, deployed via FastAPI to extract customer sentiment.
- Applied experimental design & cross-validation techniques to validate model performance, drive product insights.

Database Analyst, *Accenture Solutions Pvt. Ltd.* — *June 2021 - Aug 2022 | Hyderabad, India*

Client Data Analysis & Data Warehousing - Regeneron Pharmaceuticals Inc.

Managed large healthcare data systems by automating pipelines, maintaining cloud **data warehouse**s, and creating dashboards.

- Designed data pipelines using AWS Glue and Python to process over 30M healthcare records into Redshift and Athena.
- Optimized data warehouse performance via schema redesign, reducing query latency and improving stakeholder reporting.
- Created Tableau and Power BI dashboards for internal stakeholders, increasing data accessibility across teams.

Research Assistant - Statistics, *Rutgers University - NSF* — *March 2024 - July 2024 | Jacksonville, Florida*

Built an AI-based self-driving car system using **computer vision and sensor fusion** for lane detection and obstacle recognition.

- Scraped and analyzed 500K+ legal profiles using Puppeteer; conducted clustering, regression, and anomaly detection.
- Designed experiments and statistical models to extract workforce trends and support resource allocation policies.
- Applied regression, clustering, and time-series forecasting to uncover profession-level insights.

Researcher - Image Processing & Deep Learning, *Andhra University* — *Jan 2021 - May 2021 | Visakhapatnam, India*

Built an AI-based self-driving car system using **computer vision and sensor fusion** for lane detection and obstacle recognition.

- Built autonomous driving models using YOLO, Faster R-CNN, and CNNs with multi-sensor (LiDAR, GPS) input fusion.
- Deployed real-time testing and simulation backend using Unity3D and Flask for safety and navigation benchmarking.[Research]

TECHNICAL SKILLS

Programming: Python, R, Scala, SQL. **Libraries:** Scikit-learn, PyTorch, TensorFlow, FastAPI, Hugging Face, AutoGen.
Machine Learning: Ensemble Methods, Classification models, Gridsearching, Clustering, Regression Analysis, LightGBM.
Database & Cloud: Docker, Kafka, AWS(S3, Glue, RDS, Lambda), GCP, ETL, BigQuery, MongoDB, Langchain, MLOps.
Focus Areas: RAG Pipelines, LLM's, NLP, Experimentation Science, Pricing Analytics, ROI Modeling, Real-time AI.

EDUCATION

Rutgers University, *Masters in Data Science* — *Sept 2022 - May 2024 | New Brunswick, NJ*
Relevant Coursework: Statistics: Statistical Modeling and Computing, Stat Learning; Neural Networks, Machine Learning, Data Structures and Algorithms(Python), Data Wrangling (R), NLP, Data Mining. **Projects Overview:** Portfolio.

Andhra University, *B.Tech in Electrical and Electronics Engineering* — *June 2017 - June 2021 | Visakhapatnam, India*
Relevant Coursework: Python, Data Structures & Algorithms, Database Management, Operating Systems, System Design.

PROJECTS

Bayesian Classification - [Bayesian Networks, PyMC3, Informative Prior Distributions] Developed Bayesian logistic regression models using PyMC3 to predict diabetes status, incorporating informative priors. Evaluated performance with recall and f1-score metrics, comparing outcomes between uniform and normal distribution priors.

Language Identification - [Naive Bayes, Logistic Regression, BERT, GloVe embeddings] Worked on identifying languages using Naive Bayes, Logistic Regression, and DistilBERT, incorporating n-grams analysis achieving 96% accuracy.

Twitter Search Application - [Database Management, Python, Sorting techniques, Search Engine Optimization] Created a user query-responsive tweet retrieval system by integrating Postgres for user data management and MongoDB for tweet storage. Optimized data storage and retrieval efficiency by 35% while implementing caching strategies for system performance.

Auto Insurance Fraud Detection - [Python, Data Visualization, ML models - Random Forests, KNN, SVM, Decision Trees, XGBoost, Regression] Executed diverse ML strategies and time series analysis for auto insurance fraud, prioritizing reduced false negatives. Chose K-Nearest Neighbors (KNN) with an impressive 97% recall.